

ZOIKOSUITE

Microservices Specification Pack

Governed Business Operations Intelligence Platform

Classification

CONFIDENTIAL — INTERNAL STRATEGIC DOCUMENT

Standard

Enterprise SaaS · Engineering Specification

Control Targets

ISO 27001 · SOC 2 Type II · GDPR · CCPA-aligned architecture posture

Architecture Style

Governance-First · Event-Driven · API-First · Multi-Entity · Multi-Jurisdiction

Version

1.1 — Sovereign Microservices Specification Pack Refined

Document

03 of 06 in the ZoikoSuite Architecture Series

ARCHITECTURE SERIES

01 Sovereign Back-End Architecture

02 System Architecture Diagram Pack

03 Microservices Specification Pack (*this document*)

04 Data Model / ERD Pack

05 Security Architecture Specification

06 Engineering Build Blueprint

01 · PURPOSE OF THIS DOCUMENT

This document defines the **service-level execution architecture** of ZoikoSuite.

It specifies:

- the platform service landscape
- service responsibilities and bounded contexts
- authoritative data ownership

- canonical APIs and interface expectations
- event production and event consumption responsibilities
- governance dependencies
- evidence obligations
- scaling and deployment characteristics
- resilience, idempotency, and failure-handling rules
- commercial extensibility patterns required for enterprise growth

If Document 01 defines the doctrine, and Document 02 defines the topology, this document defines the **service contract model** the platform must be built against.

This is not conceptual guidance.

It is the **build-grade microservices blueprint** for ZoikoSuite.

02 · EXECUTIVE SUMMARY — THE SERVICE MODEL AS THE REVENUE ENGINE

ZoikoSuite services are not treated as code containers. They are treated as **value-producing platform assets**.

The service model is designed to achieve five strategic outcomes simultaneously:

- **Architectural agility** — new jurisdictions can be introduced without re-platforming core services
- **Revenue integrity** — financial leakage, duplicate execution, and workflow blind spots are structurally minimized
- **Enterprise elasticity** — tenant, entity, and jurisdiction complexity can scale without architectural drift
- **Audit defensibility** — every governed action produces evidential trace
- **Commercial extensibility** — third-party ecosystems can connect without compromising source truth or governance

The architecture therefore treats service design as a commercial instrument, not merely a technical decomposition.

03 · MICROSERVICES DESIGN DOCTRINE

ZoikoSuite does not use microservices for fashion, abstraction theater, or unnecessary complexity.

A service exists only where it creates one or more of the following:

- bounded ownership
- domain isolation
- independent scalability
- controlled blast radius
- governance enforcement
- event-driven decoupling
- deployment flexibility
- revenue-enabling extensibility

3.1 Bounded Context Before Technical Separation

A service exists because it owns a coherent domain truth, not because a function sounds architecturally important.

3.2 Governance Is Centralized, Not Reimplemented

Business domains do not self-authorize material actions. Governance, authorization, obligations, and evidence policies are enforced by shared platform services.

3.3 Source Truth Is Singular

Each material object has one authoritative owning service. All downstream representations are derived.

3.4 Events Are Preferred for Business Propagation

Services publish facts and subscribe to facts. They do not rely on brittle synchronous chaining to maintain platform truth.

3.5 APIs Are Contractual

Every service exposes explicit, versioned interfaces. Hidden coupling is prohibited.

3.6 Not Everything Should Be a Microservice

Some capabilities belong as platform services, internal libraries, or tightly coupled bounded service families. Artificial fragmentation is prohibited.

3.7 Idempotency Is a Platform Standard

Every state-changing API and every event consumer must be idempotent.

This is mandatory for:

- payment initiation
- payroll execution
- journal posting
- filing submission
- contract execution
- approval action handling

Duplicate execution risk is architecturally unacceptable.

3.8 Observability Is a Readiness Criterion

No service is production-ready unless it exposes:

- structured logs
- OpenTelemetry-compatible traces
- health probes
- service-level metrics
- correlation IDs
- alertable failure states

3.9 Governance Latency Must Remain Operationally Viable

Governance enforcement must not become a bottleneck. For latency-sensitive Tier 0 services, policy evaluation should support high-speed execution patterns, including sidecar or local decision-cache strategies where safe and auditable.

3.10 Failure Must Be Controlled, Not Improvised

Each service must define whether it fails:

- closed
- degraded
- asynchronously retried
- compensating via saga
- non-blocking to source execution

Undefined failure behavior is prohibited.

04 · SERVICE LANDSCAPE OVERVIEW

ZoikoSuite services are organized into **six service classes**.

Service Class	Purpose	Strategic Value
Governance Platform Services	Shared non-bypassable control services	Core moat
Core Domain Services	Finance, payroll, HR, legal, tax, compliance, commercial execution	Core product value
Evidence & Audit Services	Audit records, document lineage, workflow history, evidence packaging	Audit-readiness monetization
Platform Foundation Services	Identity, tenant/entity context, secrets, configuration, notifications, search	Platform survivability
Intelligence & Reporting Services	Risk scoring, anomaly detection, forecasting, reporting	Decision advantage
Integration & Extensibility Services	Safe external connectivity to banks, ERPs, HRIS, filing interfaces, e-signature, ecosystems	Onboarding speed, retention, growth

05 · SERVICE CATALOGUE

5.1 Governance Platform Services

1. Policy Service
2. Jurisdiction Rules Service
3. Authorization Service
4. Workflow & Approvals Service
5. Obligations Service
6. Evidence Requirements Service
7. Governance Decision Log Service

5.2 Identity, Scope & Foundation Services

8. Identity Context Service
9. Tenant & Entity Registry Service
10. Delegated Authority Service
11. Access Control Service
12. Secret Vault Integration Service

- 13. Configuration & Feature Flag Service
- 14. Notification Service
- 15. Search Index Service

5.3 Finance Services

- 16. General Ledger Service
- 17. Accounts Receivable Service
- 18. Accounts Payable Service
- 19. Treasury & Cash Position Service
- 20. Bank Reconciliation Service
- 21. Intercompany Accounting Service
- 22. Consolidation Service
- 23. Financial Close Service

5.4 Payroll & Workforce Services

- 24. Payroll Run Service
- 25. Compensation Service
- 26. Benefits Service
- 27. Payroll Tax Service
- 28. Payroll Exceptions Service
- 29. Employee Master Service
- 30. Employment Contracts Service
- 31. Leave & Absence Service
- 32. Org Structure Service
- 33. Performance & Review Service
- 34. Offboarding & Termination Service
- 35. Workforce Compliance Service

5.5 Legal, Corporate & Commercial Services

- 36. Contract Lifecycle Service
- 37. Clause & Template Service
- 38. Obligation Tracking Service
- 39. Board Resolutions Service
- 40. Corporate Actions Service
- 41. Counterparty Management Service
- 42. Procurement Workflow Service
- 43. Purchase Request Service
- 44. Purchase Order Service
- 45. Vendor Due Diligence Service
- 46. Invoice Approval Service
- 47. Spend Controls Service

5.6 Tax & Compliance Services

- 48. Tax Rules Service
- 49. Tax Determination Service
- 50. VAT / GST Service
- 51. Corporate Tax Estimation Service
- 52. Withholding Tax Service
- 53. Filing Preparation Service
- 54. Compliance Status Service
- 55. Filing Tracker Service
- 56. Exception & Escalation Service

5.7 Evidence, Audit & Utility Services

- 57. Audit Event Store Service
- 58. Document Vault Service
- 59. Workflow History Service
- 60. Evidence Manifest Service

5.8 Intelligence & Reporting Services

- 61. Anomaly Detection Service
- 62. Forecasting Service
- 63. Compliance Risk Scoring Service
- 64. Reconciliation Intelligence Service
- 65. Decision Support Service
- 66. Reporting Orchestration Service

5.9 Integration & Extensibility Services

- 67. Connectivity & API Bridge Service
- 68. Banking Connector Service
- 69. HRIS Connector Service
- 70. Tax Authority Interface Service
- 71. E-Signature Integration Service
- 72. External Data Feed Service

06 · PLATFORM-CRITICAL SERVICES (TIER 0)

The following services are **Tier 0** and must exist before broad functional expansion:

- Identity Context Service

- Tenant & Entity Registry Service
- Secret Vault Integration Service
- Policy Service
- Jurisdiction Rules Service
- Authorization Service
- Workflow & Approvals Service
- Obligations Service
- Governance Decision Log Service
- Audit Event Store Service
- Configuration & Feature Flag Service

Without these, ZoikoSuite may run code, but it will not operate as a governed platform.

07 · SERVICE SPECIFICATION TEMPLATE

Every service specification in ZoikoSuite must include:

- service name
- service class
- business purpose
- owned objects
- authoritative data boundary
- inbound APIs
- outbound APIs
- published events
- consumed events
- governance dependencies
- evidence obligations
- idempotency requirements
- scaling characteristics
- resilience and failure mode expectations
- deployment criticality tier

The sections below apply this structure to the most important services first.

08 · GOVERNANCE PLATFORM SERVICE SPECIFICATIONS

8.1 Policy Service

Service Class

Governance Platform Service

Purpose

Maintains and evaluates business, financial, legal, and internal control policies applicable to material actions.

Owns

- policy definitions
- policy versions
- effective dates
- policy scopes
- approval thresholds
- signatory matrices
- SoD rule sets
- spend control rules

Authoritative Boundary

Sole source of truth for platform policy definitions.

Inbound APIs

- get applicable policy set
- evaluate policy against action context
- retrieve policy version history
- validate threshold applicability

Published Events

- `policy.created`
- `policy.updated`
- `policy.version.activated`
- `policy.rule.retired`

Consumed Events

- `entity.created`
- `role.updated`
- `authority.delegated`

Evidence Obligations

- preserve every policy version
- preserve effective-dated activation
- preserve evaluation basis for governed decisions

Idempotency Requirement

Evaluation endpoints must be safely repeatable.

Scaling Characteristics

- read-heavy
- cache-accelerated
- low write frequency
- high execution criticality

Critical Constraint

Policy decisions may be cached for performance, but authoritative rule source remains centralized and auditable.

8.2 Jurisdiction Rules Service

Purpose

Maintains jurisdiction-specific payroll, tax, employment, filing, and retention rules used at runtime.

Owns

- jurisdiction profiles
- rule libraries
- effective-dated rule sets
- entity-jurisdiction applicability mappings
- compliance calendar logic
- external rule references
- legal drift indicators

Inbound APIs

- resolve jurisdiction set
- fetch runtime rule pack
- get effective rule by date
- validate action against jurisdiction requirements

Published Events

- `jurisdiction.rule.updated`
- `jurisdiction.rule.activated`
- `jurisdiction.calendar.changed`
- `legal.drift.detected`

Consumed Events

- `entity.created`
- `entity.jurisdiction.changed`
- external regulatory feed changes

Evidence Obligations

- record rule basis used in every governed action
- preserve historical rule state for replay and audit

Critical Enhancement

Supports **Legal Drift Detection** — identifies when external regulatory updates create divergence between stored platform rules and applicable legal reality.

Critical Constraint

Historical actions must always be explainable against the rule set active at time of execution.

8.3 Authorization Service

Purpose

Determines whether a principal may execute a specific action in a given tenant, entity, jurisdiction, and workflow state.

Owns

- runtime authorization decisions
- ABAC decision logic
- RBAC mappings
- SoD evaluation results
- action permission rules

Inbound APIs

- evaluate action authorization
- validate entity scope
- validate SoD conflicts
- evaluate delegated access
- retrieve authorization rationale

Published Events

- `authorization.granted`
- `authorization.denied`
- `sod.violation.detected`

Consumed Events

- `role.assigned`
- `authority.delegated`
- `employment.changed`
- `entity.scope.updated`

Evidence Obligations

- every decision logged with actor, action, basis, and outcome
- denials must be evidentially retrievable

Idempotency Requirement

Repeat evaluation must not create conflicting side effects.

Scaling Characteristics

- extremely high call volume
- latency-sensitive
- horizontally scalable
- Tier 0 criticality

Critical Constraint

No material action executes without an authorization decision artifact.

8.4 Workflow & Approvals Service

Purpose

Orchestrates approval chains for actions requiring one or more approval stages before execution.

Owns

- workflow definitions
- approval states
- escalation logic
- delegation paths
- approval deadlines

Inbound APIs

- create approval workflow
- resolve next approver
- submit approval action
- escalate pending workflow
- cancel workflow

Published Events

- `workflow.started`
- `approval.granted`
- `approval.rejected`
- `workflow.escalated`
- `workflow.completed`

Consumed Events

- `authorization.denied`
- `policy.updated`
- `authority.delegated`

Evidence Obligations

- preserve every workflow transition
- preserve approver identity and outcome
- preserve rationale where required

Idempotency Requirement

Duplicate approval submission must not create double-state transition.

Critical Constraint

Approval workflows extend authorization. They do not replace it.

8.5 Obligations Service

Purpose

Maintains active statutory, regulatory, contractual, and internal policy obligations linked to operational actions.

Owns

- obligation definitions
- obligation state
- due dates
- obligation-to-entity mappings
- obligation-to-workflow mappings
- source-origin references

Published Events

- `obligation.created`
- `obligation.updated`
- `obligation.overdue`
- `obligation.closed`

Critical Enhancement

Supports **Atomic Linking** — every obligation must be able to point to the originating source, including specific contract clause, filing rule, policy mandate, or jurisdictional rule reference.

Critical Constraint

Every obligation must be entity-bound and jurisdiction-bound.

8.6 Evidence Requirements Service

Purpose

Determines what supporting evidence must exist before an action may be completed.

Owns

- evidence preconditions
- document requirements
- signature requirements
- supporting artifact rules
- evidence sufficiency logic

Published Events

- `evidence.requirement.missing`
- `evidence.requirement.satisfied`

Critical Constraint

No finalization path may skip required evidence states.

8.7 Governance Decision Log Service

Purpose

Captures every governance evaluation as immutable evidence.

Owns

- decision records
- evaluation context
- outcome records
- rule references
- correlation IDs

Published Events

- `governance.decision.recorded`

Critical Constraint

Every governance decision must be queryable by audit, actor, entity, action, rule basis, and time range.

09 · IDENTITY, SCOPE & FOUNDATION SERVICES

9.1 Identity Context Service

Purpose

Resolves authenticated principal, session trust state, tenant context, entity scope, and delegated authority for every request.

Published Events

- `identity.context.resolved`
- `session.risk.changed`

Critical Constraint

No downstream service may infer identity context independently.

9.2 Tenant & Entity Registry Service

Purpose

Maintains authoritative structure of tenants, legal entities, hierarchies, reporting relationships, jurisdiction associations, fiscal defaults, and currency defaults.

Published Events

- `tenant.created`
- `entity.created`
- `entity.updated`
- `entity.hierarchy.changed`
- `entity.jurisdiction.changed`

Critical Constraint

Every material domain record must resolve back to an authoritative entity reference from this service.

9.3 Delegated Authority Service

Purpose

Maintains time-bound, scope-bound, approval-bound delegated authority chains.

Published Events

- `authority.delegated`
- `authority.revoked`
- `authority.expired`

Critical Constraint

Delegated authority must never exceed the delegator's own authority.

9.4 Access Control Service

Purpose

Maintains role catalogues, permission bundles, and policy-linked access groupings.

Published Events

- `role.created`
- `role.updated`
- `permission.bundle.updated`

9.5 Secret Vault Integration Service

Purpose

Provides secure brokering and controlled retrieval for sensitive credentials, bank tokens, signing keys, integration secrets, and encryption material references.

Owns

- vault integration policy
- secret access brokering
- secret lease metadata
- access audit references

Published Events

- `secret.access.requested`
- `secret.access.granted`
- `secret.rotation.completed`

Critical Constraint

No service may store long-lived sensitive credentials in local configuration or source code.

9.6 Configuration & Feature Flag Service

Purpose

Owns runtime configuration, rollout controls, and environment-aware feature flags.

Critical Constraint

Configuration may tune service behavior, but must never be used to bypass governance doctrine.

9.7 Notification Service

Purpose

Handles governed notifications for workflows, deadlines, escalations, approvals, and status changes.

Published Events

- `notification.sent`
- `notification.failed`

Critical Constraint

Notification failure must not collapse source operational workflows.

9.8 Search Index Service

Purpose

Provides fast retrieval of documents, obligations, clauses, and evidence metadata.

Critical Constraint

Search is derivative, never authoritative.

10 · FINANCE DOMAIN SERVICE SPECIFICATIONS

10.1 General Ledger Service

Purpose

Authoritative owner of journalized financial postings and ledger state.

Owns

- journal headers
- journal lines
- posting state
- fiscal period linkage
- account references
- posting validation lifecycle

Published Events

- `journal.created`
- `journal.validated`
- `journal.posted`
- `journal.reversed`

Consumed Events

- `payment.initiated`
- `invoice.approved`

- `payroll.run.completed`
- `intercompany.entry.posted`

Evidence Obligations

- preserve actor
- preserve approval references
- preserve period state
- preserve validation lineage

Critical Enhancement

Supports **Tri-Phase Commit States**:

- Pending
- Validated
- Finalized

This enables real-time control, intelligence checks, and evidential validation before immutable finalization.

Critical Constraint

No finalized journal may be hard-edited. Corrections occur only through reversal or adjustment.

10.2 Accounts Receivable Service

Purpose

Owns customer invoicing, receivable state, collection status, and settlement linkage.

Published Events

- `invoice.issued`
- `invoice.sent`
- `receivable.overdue`
- `payment.received`

Critical Constraint

Receivable state must reconcile to authoritative ledger truth.

10.3 Accounts Payable Service

Purpose

Owns vendor invoice intake, liability-side invoice lifecycle, and payment readiness state.

Published Events

- `vendor.invoice.received`

- `vendor.invoice.validated`
- `vendor.invoice.approved`
- `payment.requested`

Critical Constraint

No payable may proceed to payment initiation without approval-state and evidence-state validation.

10.4 Treasury & Cash Position Service

Purpose

Provides governed liquidity visibility across entities and bank accounts.

Published Events

- `cash.position.updated`
- `liquidity.threshold.breached`
- `effective.cash.position.updated`

Critical Enhancement

Supports **Intra-day Liquidity Forecasting** and computes **Effective Available Cash**, net of:

- pending AP commitments
- payroll obligations
- tax liabilities
- reserved approvals not yet settled

This is critical for high-growth operating resilience.

10.5 Bank Reconciliation Service

Purpose

Owns statement matching, reconciliation state, exception queues, and reconciliation evidence.

Published Events

- `statement.ingested`
- `reconciliation.matched`
- `reconciliation.exception.raised`
- `reconciliation.completed`

Critical Constraint

Suggested matches may be intelligence-assisted, but final reconciliation state remains governed and evidential.

10.6 Intercompany Accounting Service

Purpose

Owns governed intercompany entries, matching logic, and balancing integrity.

Published Events

- `intercompany.entry.created`
- `intercompany.entry.posted`
- `intercompany.mismatch.detected`

Critical Constraint

Intercompany activity must never be collapsed into single-entity truth.

10.7 Consolidation Service

Purpose

Produces multi-entity consolidated financial views based on source domain truth.

Published Events

- `consolidation.run.started`
- `consolidation.completed`
- `consolidation.exception.detected`

Critical Constraint

Consolidated outputs are derived views, not replacements for source entity truth.

10.8 Financial Close Service

Purpose

Owns period close orchestration, readiness checks, lock state, and close evidence.

Published Events

- `period.close.started`
- `period.close.blocked`
- `period.closed`

Critical Constraint

Period close must generate immutable close-state evidence.

11 · PAYROLL & WORKFORCE DOMAIN SERVICE SPECIFICATIONS

11.1 Payroll Run Service

Purpose

Orchestrates end-to-end payroll execution for a given entity, cycle, and jurisdiction set.

Published Events

- `payroll.run.initiated`
- `payroll.run.calculated`
- `payroll.run.completed`
- `payroll.run.blocked`

Critical Constraint

Payroll runs must be snapshot-based and immutable after finalization.

11.2 Compensation Service

Purpose

Owns salary, bonus, variable pay, and compensation package definitions.

Published Events

- `compensation.updated`
- `bonus.approved`
- `compensation.effective.changed`

11.3 Benefits Service

Purpose

Owns benefit enrollments, plan participation, and benefit-linked payroll impacts.

Published Events

- `benefit.enrolled`
- `benefit.changed`
- `benefit.terminated`

11.4 Payroll Tax Service

Purpose

Calculates payroll tax obligations using jurisdiction-aware rules.

Published Events

- `payroll.tax.calculated`

- `payroll.tax.exception.detected`

Critical Enhancement

Implements **Multi-Tax-Engine Abstraction** so local providers, external tax engines, or government interfaces may be plugged in without rewriting payroll core logic.

Critical Constraint

Every calculation must preserve rule basis and provider basis used at execution time.

11.5 Payroll Exceptions Service

Purpose

Captures anomalies, conflicts, insufficient evidence states, and release blockers.

Published Events

- `payroll.exception.raised`
- `payroll.exception.resolved`

11.6 Employee Master Service

Purpose

Authoritative owner of workforce identity and lifecycle state.

Published Events

- `employee.created`
- `employee.hired`
- `employee.status.changed`
- `employee.terminated`

Critical Constraint

Owns employee core truth, but not all derived workforce artifacts.

11.7 Employment Contracts Service

Purpose

Owns employment contract issuance, amendment, and version lineage.

Published Events

- `employment.contract.issued`
- `employment.contract.amended`
- `employment.contract.terminated`

Critical Constraint

No contract amendment may overwrite prior contract state.

11.8 Leave & Absence Service

Purpose

Owns leave requests, balances, approvals, and jurisdiction-sensitive leave logic.

Published Events

- `leave.requested`
- `leave.approved`
- `leave.rejected`
- `leave.balance.updated`

11.9 Org Structure Service

Purpose

Maintains departments, positions, reporting lines, and organizational placement.

Published Events

- `position.created`
- `employee.assigned`
- `org.structure.changed`

11.10 Performance & Review Service

Purpose

Owns review cycles, review records, and governed performance artifacts.

Published Events

- `review.created`
- `review.completed`

11.11 Offboarding & Termination Service

Purpose

Owns governed offboarding and termination workflows.

Published Events

- `termination.initiated`
- `termination.approved`

- `employee.terminated`
- `offboarding.completed`

Critical Constraint

Termination workflows must validate jurisdictional employment law before execution.

11.12 Workforce Compliance Service

Purpose

Monitors employment-law-sensitive obligations and workforce compliance sufficiency.

Published Events

- `workforce.compliance.gap.detected`
- `workforce.compliance.resolved`

12 · LEGAL, CORPORATE & COMMERCIAL DOMAIN SERVICE SPECIFICATIONS

12.1 Contract Lifecycle Service

Purpose

Owns contract drafting, negotiation, execution, renewal, and lifecycle status.

Published Events

- `contract.drafted`
- `contract.submitted`
- `contract.executed`
- `contract.renewal.due`
- `contract.expired`

Critical Constraint

No contract becomes active without governed approval and signatory validation.

12.2 Clause & Template Service

Purpose

Owns standard clause libraries, approved templates, and clause-governance controls.

Published Events

- `clause.template.updated`
- `template.approved`

12.3 Obligation Tracking Service

Purpose

Extracts, stores, and manages contract-linked obligations as machine-trackable objects.

Published Events

- `contract.obligation.created`
- `contract.obligation.overdue`

Critical Constraint

Must support atomic reference back to source contract clause and version.

12.4 Board Resolutions Service

Purpose

Owns board-resolution draft, approval, execution, and archival state.

Published Events

- `resolution.drafted`
- `resolution.approved`
- `resolution.executed`

12.5 Corporate Actions Service

Purpose

Owns governed corporate actions such as officer changes, filings, and board-authorized entity events.

Published Events

- `corporate.action.initiated`
- `corporate.action.completed`
- `entity.filing.submitted`

12.6 Counterparty Management Service

Purpose

Owns legal counterparties, validation state, and risk metadata.

Published Events

- `counterparty.created`
- `counterparty.validated`
- `counterparty.risk.changed`

12.7 Procurement Workflow Service

Purpose

Owns procurement orchestration and governed spend routing.

Published Events

- `procurement.requested`
- `procurement.approval.started`
- `procurement.completed`

12.8 Purchase Request Service

Purpose

Owns purchase-request objects and lifecycle before order issuance.

Published Events

- `purchase.request.created`
- `purchase.request.approved`
- `purchase.request.rejected`

12.9 Purchase Order Service

Purpose

Owns purchase-order issuance, amendment, and fulfillment-linked state.

Published Events

- `purchase.order.issued`
- `purchase.order.amended`
- `purchase.order.closed`

12.10 Vendor Due Diligence Service

Purpose

Owns vendor validation, onboarding checks, sanctions/risk state, and due-diligence evidence.

Published Events

- `vendor.dd.started`
- `vendor.dd.completed`
- `vendor.dd.failed`

12.11 Invoice Approval Service

Purpose

Owns governed invoice approval state prior to payable release.

Published Events

- `invoice.approval.started`
- `invoice.approved`
- `invoice.rejected`

12.12 Spend Controls Service

Purpose

Owns spend-control enforcement logic and consumption tracking against policy.

Published Events

- `spend.threshold.breached`
- `spend.block.applied`

13 · TAX & COMPLIANCE SERVICE SPECIFICATIONS

13.1 Tax Rules Service

Purpose

Maintains structured tax-rule libraries and effective-dated tax logic references.

Published Events

- `tax.rule.updated`
- `tax.rule.activated`

13.2 Tax Determination Service

Purpose

Determines applicable tax treatment for governed financial or payroll actions.

Published Events

- `tax.determined`
- `tax.determination.failed`

13.3 VAT / GST Service

Purpose

Calculates and tracks indirect tax obligations for transactional flows.

Published Events

- `vat.calculated`
- `gst.calculated`
- `indirect.tax.exception.detected`

13.4 Corporate Tax Estimation Service

Purpose

Produces governed estimated tax positions for entity and group contexts.

Published Events

- `corporate.tax.estimated`

13.5 Withholding Tax Service

Purpose

Calculates and tracks withholding obligations tied to payment flows.

Published Events

- `withholding.tax.calculated`

13.6 Filing Preparation Service

Purpose

Owns pre-submission filing assembly and evidence completeness validation.

Published Events

- `filing.prepared`
- `filing.blocked`
- `filing.ready.for.submission`

13.7 Compliance Status Service

Purpose

Produces explainable status for obligations, controls, and compliance readiness.

Published Events

- `compliance.status.changed`
- `compliance.gap.detected`

13.8 Filing Tracker Service

Purpose

Tracks filing lifecycle from due-date planning through submission and confirmation.

Published Events

- `filing.due`
- `filing.submitted`
- `filing.confirmed`
- `filing.overdue`

13.9 Exception & Escalation Service

Purpose

Owns governed exception cases and escalation routing.

Published Events

- `exception.created`
- `exception.escalated`
- `exception.closed`

14 · EVIDENCE, AUDIT & UTILITY SERVICE SPECIFICATIONS

14.1 Audit Event Store Service

Purpose

Stores append-only, audit-grade event records correlated to source actions.

Critical Constraint

Records must be immutable and queryable by actor, entity, action, workflow, or time range.

14.2 Document Vault Service

Purpose

Stores critical documents with version lineage, access lineage, integrity controls, and retention policy.

14.3 Workflow History Service

Purpose

Stores complete workflow transition history for approvals, escalations, and action states.

14.4 Evidence Manifest Service

Purpose

Builds structured evidence sets for audit, regulator, legal-discovery, and compliance-review scenarios.

Published Events

- `evidence.manifest.generated`

Commercial Note

This service is strategically monetizable as a premium “Instant Audit” capability.

15 · INTELLIGENCE & REPORTING SERVICE SPECIFICATIONS

15.1 Anomaly Detection Service

Purpose

Detects unusual patterns across finance, payroll, compliance, and workflow data.

Commercial Positioning

May be externally positioned as **Revenue Integrity** when used to identify missed billables, leakage, overpayments, or abnormal spend patterns.

Critical Constraint

May flag and score. May not mutate operational truth.

15.2 Forecasting Service

Purpose

Produces forward-looking tax, payroll, liability, and operational forecasts.

15.3 Compliance Risk Scoring Service

Purpose

Scores obligations, filings, and control exposures by severity, proximity, and entity impact.

15.4 Reconciliation Intelligence Service

Purpose

Provides recommended matches, exception prioritization, and reconciliation assistance.

15.5 Decision Support Service

Purpose

Provides governed recommendations at approval and oversight points.

15.6 Reporting Orchestration Service

Purpose

Coordinates executive, board, operational, and regulatory reporting jobs from derived stores.

Critical Constraint

Reporting services are read-derived only. They do not own source truth.

16 · INTEGRATION & EXTENSIBILITY SERVICES

16.1 Connectivity & API Bridge Service

Purpose

Provides controlled ingestion and normalization of external platform data into governed internal flows.

Strategic Value

Reduces onboarding friction, speeds up enterprise migration, and improves retention.

Critical Constraint

Imported data must preserve provenance and must never bypass governance before operational use.

16.2 Banking Connector Service

Purpose

Handles payment rails, bank statement ingestion, and treasury connectivity.

16.3 HRIS Connector Service

Purpose

Supports coexistence with retained HRIS platforms during phased adoption or hybrid operating models.

16.4 Tax Authority Interface Service

Purpose

Supports filing submission, status confirmation, and external compliance channel integration.

16.5 E-Signature Integration Service

Purpose

Supports governed external execution of contracts, board resolutions, and legal artifacts.

16.6 External Data Feed Service

Purpose

Ingests external regulatory, market, and jurisdictional data sources.

Critical Enhancement

Feeds legal drift detection and rule-change monitoring across jurisdictions.

17 · SERVICE-TO-SERVICE INTERACTION RULES

17.1 No Domain Service Self-Authorizes

Material actions must pass through governance services before execution.

17.2 No Cross-Service Source Truth Mutation

A service may not write directly into another service's authoritative store.

17.3 Events Are Preferred for Business State Propagation

Inter-service coordination should prefer domain events over brittle synchronous chains.

17.4 Synchronous Calls Are Reserved for Narrow Control Paths

Examples:

- identity-context resolution
- authorization decisions
- policy lookup
- workflow command execution

17.5 Every Material Service Must Be Entity-Aware

No service may operate on material objects without tenant, entity, and jurisdiction context.

17.6 Every Material Service Must Be Evidential

Any service changing important business state must emit events and support evidence generation.

17.7 Circuit Breaker Mandate

All inter-service calls must use circuit breakers, timeouts, retries, and dead-letter routing where applicable.

A Tier 2 failure must not collapse a Tier 0 process.

17.8 Saga Discipline for Distributed Workflows

Long-running multi-service flows must use governed saga patterns with compensating transactions where required.

Example: Hire-to-Pay or Procure-to-Pay flows spanning employee, payroll, tax, ledger, and obligations services.

18 · DATA OWNERSHIP RULES

Each material object must have one authoritative owner.

Object	Authoritative Service
Legal Entity	Tenant & Entity Registry Service
Journal Entry	General Ledger Service
Employee Profile	Employee Master Service
Employment Contract	Employment Contracts Service
Payroll Run	Payroll Run Service
Contract	Contract Lifecycle Service
Tax Rule	Tax Rules Service
Filing Status	Filing Tracker Service
Obligation	Obligations Service or Obligation Tracking Service by source class
Workflow Instance	Workflow & Approvals Service

No service may claim ownership of an object already owned elsewhere.

19 · EVENT CONTRACT RULES

Every published event must include, at minimum:

- event name
- event version
- event timestamp
- tenant ID
- legal entity ID
- jurisdiction context
- actor ID or system principal
- correlation ID
- source service
- payload schema version

Breaking schema changes require explicit version transition planning.

Event consumers must be replay-safe and idempotent.

20 · RESILIENCE, DR & GLOBAL CONTINUITY RULES

20.1 Disaster Recovery Is Architectural, Not Operational

The platform must support cross-region recovery for Tier 0 and Tier 1 services.

20.2 Global Traffic Management

Ingress routing must support regional failover and controlled traffic diversion for continuity events.

20.3 Cross-Region Replication

Required for:

- critical event logs
- audit evidence
- tenant/entity registry
- core financial and payroll continuity data
subject to jurisdictional residency rules.

20.4 Recovery Targets

RTO and RPO targets must be defined by service tier and enforced through deployment architecture.

20.5 Replay Safety

Recovery must never produce duplicate financial, payroll, or filing side effects.

21 · SCALING & DEPLOYMENT CHARACTERISTICS

High-QPS / Low-Latency Services

- Authorization Service
- Identity Context Service

- Policy Service
- Search Index Service
- Notification Service

Throughput-Heavy / Batch-Sensitive Services

- Payroll Run Service
- Bank Reconciliation Service
- Consolidation Service
- Filing Preparation Service
- Reporting Orchestration Service

Storage-Heavy / Retention-Sensitive Services

- Audit Event Store Service
- Document Vault Service
- Workflow History Service
- Evidence Manifest Service

High-Criticality / Strong-Consistency Services

- General Ledger Service
- Payroll Run Service
- Workflow & Approvals Service
- Authorization Service
- Tenant & Entity Registry Service

Deployment patterns must reflect:

- service criticality tier
- consistency requirements
- failure blast radius
- recovery expectations
- compliance obligations

22 · FAILURE MODE EXPECTATIONS

Every service specification must define:

- what happens if the service is unavailable
- whether requests fail closed or retry asynchronously
- whether events are replay-safe
- whether compensating transactions are required
- whether evidence remains recoverable

Platform rule:

- governance services fail closed
- evidence services fail safe and durable
- reporting services fail degraded, not destructive
- intelligence services fail without mutating source execution unless explicitly control-relevant

23 · SERVICE TIERS

Tier 0 — Platform Survival

Identity, authorization, policy, jurisdiction, tenant/entity, workflow, audit event store, secret vault integration

Tier 1 — Core Operational Execution

Ledger, payroll, employee master, contract lifecycle, obligations, filing tracker

Tier 2 — Platform Completeness

Benefits, reconciliation intelligence, reporting orchestration, search, notifications, connectors

Tier 3 — Optimization & Enhancement

Forecasting, decision support, advanced anomaly detection, extended integrations

This tiering governs:

- deployment priority
- uptime target
- incident severity
- rollback discipline
- staffing allocation

24 · ENGINEERING OWNERSHIP MODEL

Team	Service Families
Platform Governance Team	Policy, jurisdiction, authorization, workflow, obligations, decision log

Identity & Access Team	Identity context, access control, delegated authority, tenant/entity, secret vault
Finance Platform Team	Ledger, AR, AP, treasury, reconciliation, close, consolidation
Workforce Platform Team	Payroll, employee master, contracts, leave, org, offboarding
Legal & Commercial Team	Contract lifecycle, corporate actions, procurement, vendor DD, spend controls
Tax & Compliance Team	Tax determination, VAT/GST, filings, compliance status, exceptions
Evidence & Audit Platform Team	Audit event store, document vault, workflow history, evidence manifest
Intelligence & Reporting Team	Forecasting, risk scoring, anomaly detection, reporting orchestration
Integration Platform Team	Banking connectors, HRIS connectors, e-signature, external data feeds

Shared ownership without explicit accountability is prohibited.

25 · BUILD ORDER

Phase 0 — Foundation

- Identity Context Service
- Tenant & Entity Registry Service
- Secret Vault Integration Service

Phase 1 — Governance Spine

- Policy Service
- Jurisdiction Rules Service
- Authorization Service
- Workflow & Approvals Service
- Governance Decision Log Service
- Audit Event Store Service

Phase 2 — Revenue Engine

- General Ledger Service

- Accounts Payable Service
- Accounts Receivable Service
- Contract Lifecycle Service

Phase 3 — Workforce Engine

- Employee Master Service
- Employment Contracts Service
- Payroll Run Service
- Payroll Tax Service

Phase 4 — Compliance & Intelligence Overlay

- Obligations Service
- Filing Tracker Service
- Evidence Manifest Service
- Anomaly Detection Service

Phase 5 — Platform Maturity

- Document Vault Service
- Search Index Service
- Notification Service
- Reporting Orchestration Service
- Connector services

This build order protects architectural integrity while accelerating commercial readiness.

26 · FINAL MICROSERVICES DOCTRINE

ZoikoSuite services are separated not merely by function, but by:

- truth ownership
- governance dependency
- event responsibility
- evidential consequence
- scaling profile
- commercial value

Every service must answer four questions clearly:

1. What truth do I own?
2. What governed actions do I enable?
3. What evidence do I create or require?
4. What events do I publish for the rest of the platform?

If a service cannot answer those questions cleanly, it is not yet correctly bounded.

CTO ASSESSMENT

This refined Microservices Specification Pack brings ZoikoSuite to a true build-grade Tier 1 standard because it:

- translates doctrine into enforceable service ownership
- integrates commercial growth logic into architecture
- strengthens idempotency, resilience, and DR posture
- protects against revenue leakage and duplicate execution
- preserves governance as the non-bypassable platform spine
- creates a platform extensibility layer that speeds onboarding and reduces churn

This is where ZoikoSuite stops being a concept and becomes a disciplined enterprise system.

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